

Tests

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Solutions

Questions

Question 1: PON transmission control

How is the problem arising in a PON when a number of Optical Network Units (ONU) try to transmit simultaneously solved?

- a) The problem cannot be solved, but the ONUs are able to retransmit after a collision of their packets.
- b) The power emitted by all the ONUs is regulated in order to have the same value.

- c) The OLT controls the transmissions of the ONUs by giving the right scheduling to avoid collisions.
- d) The ONUs buffer their data until they find the fiber free for their transmission.

Question 2: Wavelength Division Multiplexing (WDM)

How is Wavelength Division Multiplexing (WDM) used in fiber-optic transmission for a PON?

- a) If we have 10 different wavelengths, we can have 10 different colors contemporarily in the same fiber.
- b) The same Optical Network Unit can transmit different packets on different wavelengths.
- c) One wavelength is used for the downstream and one for the upstream.
- d) If we have 10 different wavelengths, we can have 10 different fibers contemporarily in the same cable.

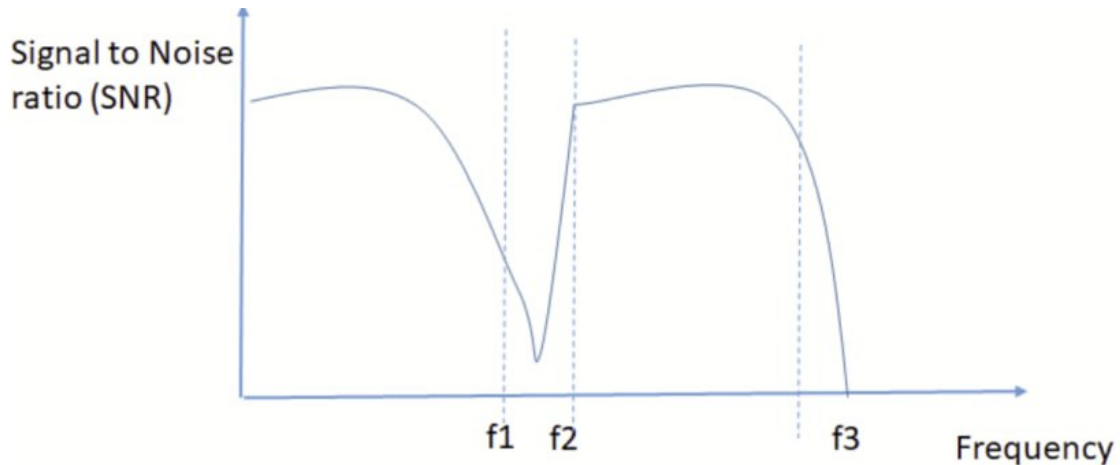
Question 3: Passive Optical Networks benefits

What are the benefits of Passive Optical Networks (PON) compared to Active Optical Networks?

- a) The infrastructure cost is lower due to simpler equipment and fewer active components.
- b) It's easier to manage thanks to the use of Ethernet-based approaches.
- c) You avoid the need for Optical Network Units at the user side.
- d) You only need to replace damaged components, not the entire system.

Question 4: ADSL channel behavior and Discrete Multitone Technology (DMT)

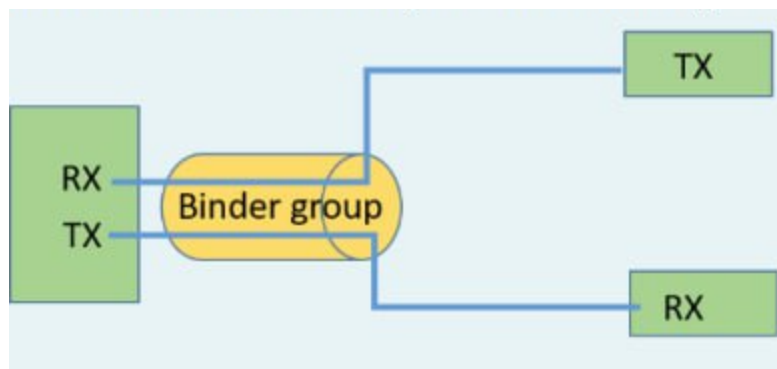
Considering an ADSL line with frequencies fully attenuated (low SNR) in certain ranges, how should a modulation based on Discrete Multitone Technology (DMT) handle these frequencies?



- a) It should include transmitting "tones" in the frequency band below f_1 and between f_2 and f_3 .
- b) It must include only transmitting "tones" in the frequency band between f_2 and f_3 .
- c) It should include transmitting "tones" only in the frequency band below f_1 .

Question 5: Cross-talk in shared cable paths

What type of cross-talk might arise when two pairs of cables share an initial path before diverging?



- a) The Near End Cross Talk.
- b) None of the cross talks may arise in this setting.
- c) Both the two kinds of cross-talks may arise.
- d) The Far End Cross Talk.

Question 6: FTTX configurations and deployment cost

Which of the FTTX configurations has a higher deployment cost?

- a) FTTB - to the building.
- b) The cost is identical, given the fact that fibers shall be deployed in this access network part.
- c) FTTC - to the curb.
- d) FTTE - to the exchange.

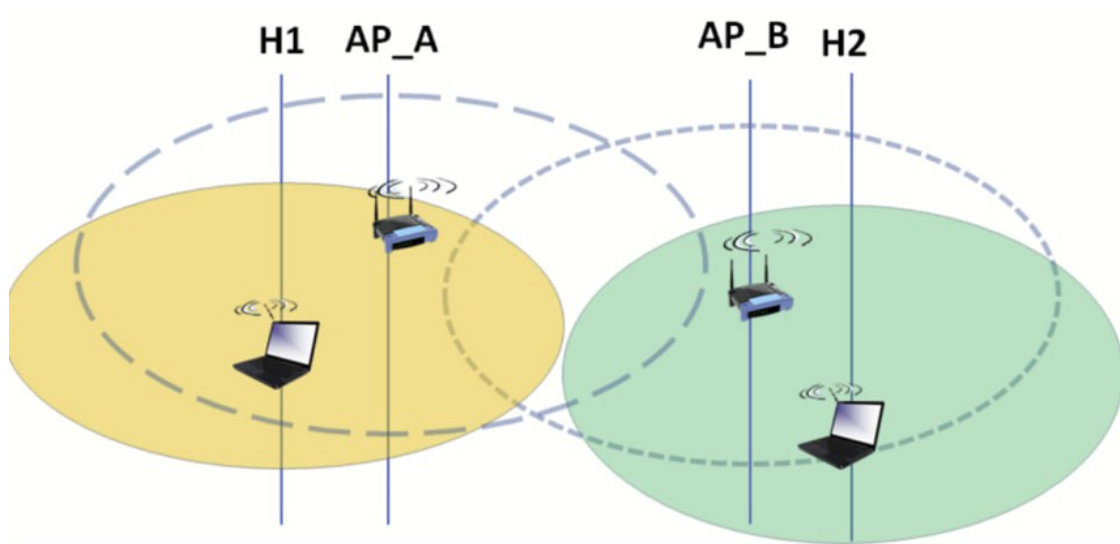
Question 7: Water filling algorithm in ADSL

When are the advantages of applying the water filling algorithm in an ADSL system most noticeable?

- a) The channel behavior both in the upstream and downstream has a flat (constant) signal-to-noise ratio.
- b) It is important to separate the telephone bandwidth from the one used by the data.
- c) It is important to split the transmission into two parts: downstream and upstream.
- d) The channel behavior both in the upstream and downstream has not a flat (constant) signal-to-noise ratio.

Question 8: Hidden terminal problem in a network with Access Points and Hosts

In a network with two Hosts (H1 and H2) and two Access Points (AP_A and AP_B) sharing the same frequency band, can H1-AP_A and H2-AP_B transmit simultaneously?



- a) We cannot have contemporary transmissions of H1-AP_A and H2-AP_B since H1 and H2 are hidden terminals.
- b) We can have contemporary bidirectional transmissions of H1-AP_A and H2-AP_B since H1 and H2 are in two separate coverage areas, and the same happens for the Access Points.
- c) We cannot have contemporary bidirectional transmissions of H1-AP_A and H2-AP_B since AP_A and AP_B have an overlapping coverage area.

Question 9: VDSL improvements

Which are the two improvements enabling the VDSL?

- a) The use of the DMT modulation in the copper cables and the full cancellation of the cross-talk noise from the CO to the end user.
- b) The use of high frequencies in the fiber optic cables and the cancellation of the interference arising in the fiber optic domain.
- c) The vectoring of the signals in the fibers cables that arrive till the end users and the positioning of the ONU close to the end users.
- d) The vectoring of the signals in the copper cables and the placement of the DSLAM (where the fiber ends) close to the end user.

Question 10: Importance of latency in future wireless systems

In the future generation wireless systems, the latency will become a key performance metric. Why is it becoming so important?

- a) Future mobile applications will use high data processing and computation.
- b) Many mobile applications will require high responsiveness in the interaction with the network.
- c) The end users may have high mobility during their connection.
- d) Many mobile applications will require high throughput (bit rate) at the end users.

Question 11: Critical impairment in wireless infrastructure

Which is the most critical impairment in a wireless infrastructure when multiple end users access the wireless link?

- a) The multipath, since we have multiple users.
- b) The path loss, since multiple users may be very distant from the base station.
- c) The interference that they may generate one with the other if the multiple access is not suitably controlled.

Question 12: Water filling algorithm in ADSL

Suppose to apply the water filling algorithm in an ADSL system. The main advantages of this approach are attained when:

- a) The channel behaviour both in the upstream and in the downstream has a flat (constant) signal to noise ratio.
- b) It is important to separate the telephone bandwidth for the one used by the data.
- c) It is important to split in two parts the transmission in the downstream and in the upstream.
- d) The channel behaviour both in the upstream and in the downstream has not a flat (constant) signal to noise ratio.

Question 13: Fiber cut in optical subnetworks

Considering two all optical subnetworks (say A and B) interconnected through an OXC. Assuming that there is a fiber cut in the subnetwork A, then:

- a) All the FDI signals are propagated from A to B.
- b) Only OTS-FDI are propagated from A to B.
- c) No FDI signals are propagated from A to B.
- d) Only ODU-FDI are propagated from A to B.

Question 14: Topologies for networks in access part

Which are the most used topologies for networks in the access part?

- a) Mesh and bus.
- b) Rings.
- c) Stars and rings.
- d) Bus.

Question 15: Best configuration in terms of expected bit rate

Which one of these configurations is the best in terms of expected bit rate?

- a) FTTC - to the curb.
- b) ADSL.
- c) FTTE - to the exchange.
- d) FTTB - to the building.

Question 16: Role of the EDGE in a network

What is the EDGE of a network and which is its role?

- a) The very core part of a network where powerful network functions are implemented.
- b) That part of the network between the access and the core with some intelligent functions.
- c) The very peripheral part of the network where only simple functions are implemented.
- d) The very peripheral part of the network where all the applications are.

Question 17: Cross-talk noise on ADSL

What is the cross-talk noise on ADSL?

- a) A noise that is generated at a generic receiver within a copper cable.
- b) The coupling of signals from a circuit to another giving rise to undesired interference.
- c) The interference of a signal in a frequency band to signals in other frequency bands of the ADSL.
- d) A noise that is generated at a generic receiver within a fiber cable.

Question 18: Evolution of the access network

The evolution of the current access network aims at:

- a) Employing fiber cables instead of copper ones.
- b) Providing an ADSL access to all the users.
- c) Using in a more capillary way the already existing infrastructure based on copper cables.
- d) Using only wireless access given its flexibility and low cost.

Question 19: Key element of ADSL success

One of the key elements of ADSL success is related to:

- a) The use of the same copper cable of the old telephone network.
 - b) The fact that no new network elements are needed at the Central Office side.
 - c) The use of the same frequency band of the old telephone network.
 - d) The use of new fiber cables in place of the old telephone ones.
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Solutions

- **Question 1:** c) The OLT controls the transmissions of the ONUs by giving the right scheduling to avoid collisions.

- **Question 2:**a) If we have 10 different wavelengths, we can have 10 different colors contemporarily in the same fiber.
- **Question 3:** a) The infrastructure cost is lower due to simpler equipment and fewer active components.
- **Question 4:** a) It should include transmitting "tones" in the frequency band below f_1 and between f_2 and f_3 .
- **Question 5:** a) The Near End Cross Talk.
- **Question 6:** a) FTTB - to the building.
- **Question 7:** d) The channel behavior both in the upstream and downstream has not a flat (constant) signal-to-noise ratio.
- **Question 8:** b) We can have contemporary bidirectional transmissions of H1-AP_A and H2-AP_B since H1 and H2 are in two separate coverage areas, and the same happens for the Access Points.
- **Question 9:** d) The vectoring of the signals in the copper cables and the placement of the DSLAM (where the fiber ends) close to the end user.
- **Question 10:** b) Many mobile applications will require high responsiveness in the interaction with the network.
- **Question 11:** c) The interference that they may generate one with the other if the multiple access is not suitably controlled.
- **Question 12:** d) The channel behaviour both in the upstream and in the downstream has not a flat (constant) signal to noise
- **Question 13:** b) Only OTS-FDI are propagated from A to B.
- **Question 14:** c) Stars and rings.
- **Question 15:** d) FTTB - to the building.
- **Question 16:** b) That part of the network between the access and the core with some intelligent functions.
- **Question 17:** b) The coupling of signals from a circuit to another giving rise to undesired interference.
- **Question 18:** a) Employing fiber cables instead of copper ones.

- **Question 19:** a) The use of the same copper cable of the old telephone network.